



What European are doing right about cybersecurity?

Lessons from ENISA and the US National Cybersecurity Strategy

Defcon 33 | Muhammad Mudassar Yamin

Whoami?

Professional Title

- Associate Professor NTNU (Norwegian University of Science and Technology)

Expertise

- Cybersecurity: System security, penetration testing, intrusion detection
- AI & Machine Learning: Applications in cybersecurity
- Cyber Range Training: Development and implementation

Certifications

- OSCE, OSCP, LPT-MASTER, CEH, CPTE, CHFI, CPTE, CISSO, CBP, CCNA-CyberOPS, KLSE

Awards

- INTERPOL Medal for Innovation
- 50+ HoF (Microsoft, TrendMicro, US DOD)

Key Projects

- Open Cyber Range, CR14 NATO CCDCOE
- Towards a common ECSC roadmap, ENISA

Publications

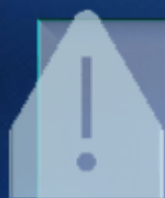
- "Cyber ranges and security testbeds: Scenarios, functions, tools and architecture." Computers & Security 88 (2020): 101636.
- "Weaponized AI for cyber attacks" (Journal of Information Security and Applications 57 (2021): 102722)
- "Modeling and executing cyber security exercise scenarios in cyber ranges " Computers & Security 116 (2022): 102635.

Contact

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The Global Cybersecurity Landscape

In today's interconnected world, cybersecurity has become **critical to national security, economic prosperity, and public safety**. The threat landscape continues to evolve with:



Global Threats

Nation-state actors, criminal syndicates, and hacktivists operating across borders



Critical Infrastructure

Increasing attacks on essential services and systems that power our societies



International Cooperation

Growing need for collaborative defense and shared responsibility

This presentation examines how Europe and the United States are approaching these challenges through their respective cybersecurity strategies and initiatives.

Europe's Approach: The ENISA Perspective

The European Union Agency for Cybersecurity (ENISA) plays a central role in Europe's cybersecurity ecosystem, focusing on:

Cybersecurity Education & Skills Development

Building human capital through structured training and competitions

European Cybersecurity Challenge (ECSC)

A flagship initiative to identify talent and create a network of specialists

Public-Private Partnerships

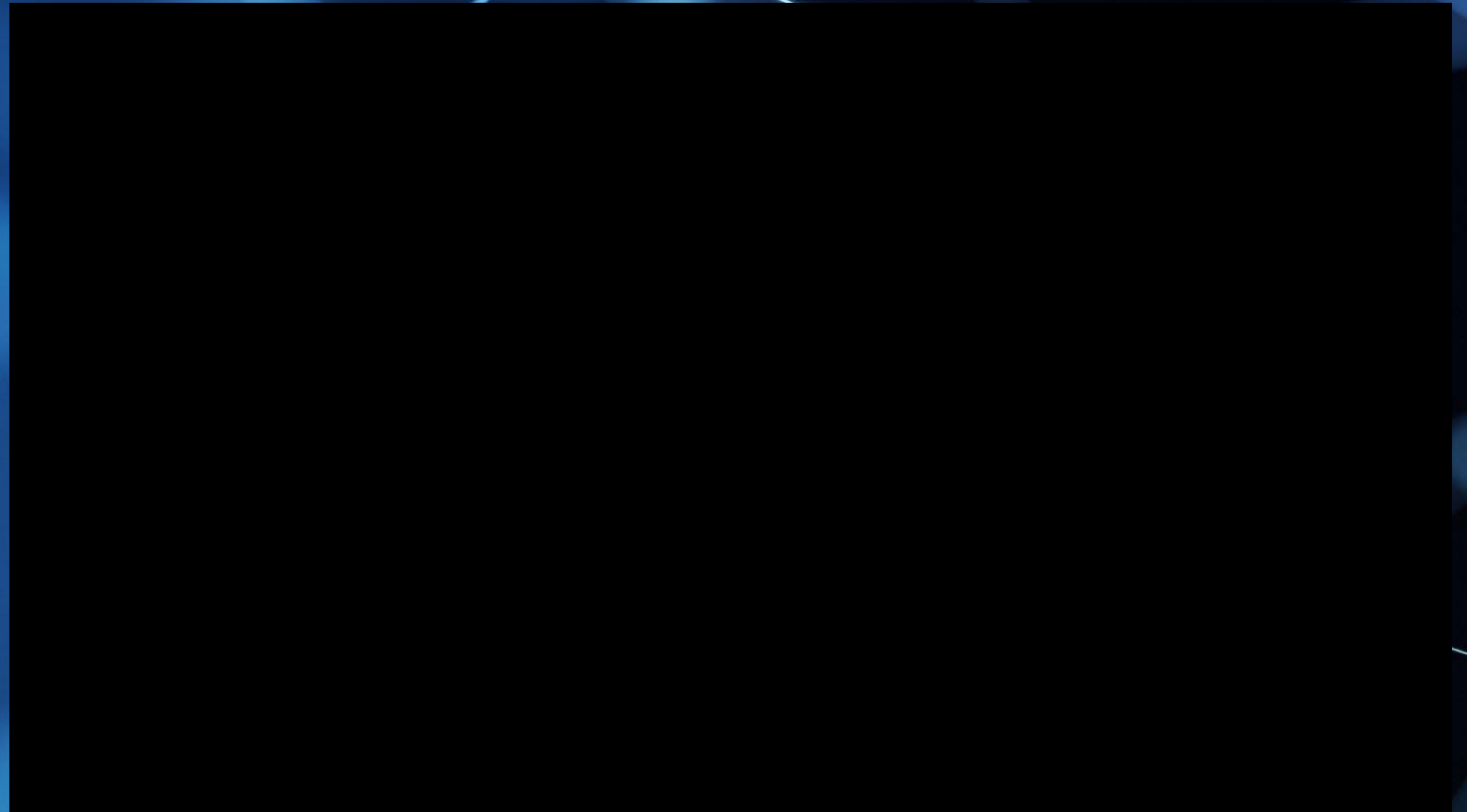
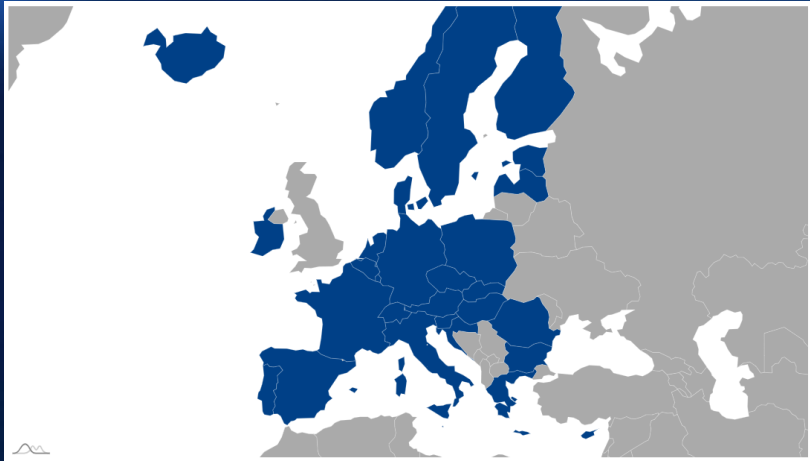
Fostering collaboration between government, industry, and academia

Standardization Across Member States

Creating common frameworks and approaches for a unified European response

Europe's Policy in Action

Building human capital through structured training and competitions



USA's Approach: National Cybersecurity Strategy

The U.S. National Cybersecurity Strategy introduces two fundamental shifts:

- Rebalancing responsibility to defend cyberspace
- Realigning incentives for long-term investments

The Five Strategic Pillars:

1. Defend Critical Infrastructure

Mandatory requirements, public-private collaboration, modernizing Federal defenses

2. Disrupt and Dismantle Threat Actors

Integrated disruption activities, intelligence sharing, countering cybercrime

3. Shape Market Forces to Drive Security and Resilience

Shifting liability for insecure software, secure IoT development, Federal procurement

4. Invest in a Resilient Future

R&D, post-quantum security, clean energy, digital identity, cyber workforce

5. Forge International Partnerships

Building coalitions, strengthening partner capacity, securing global supply chains

USA's Policy in Action

Integrated disruption activities, intelligence sharing, countering cybercrime

PRESS RELEASE

U.S. Government seizes approximately 145 criminal marketplace domains

Wednesday, June 4, 2025

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For Immediate Release

U.S. Attorney's Office, Eastern District of Virginia

ALEXANDRIA, Va. – The U.S. Attorney's Office for the Eastern District of Virginia announced today the seizure of approximately 145 darknet and traditional internet domains, and cryptocurrency funds associated with the BidenCash marketplace. The operators of the BidenCash marketplace use the platform to simplify the process of buying and selling stolen credit cards and associated personal information.

BidenCash commenced operations in March 2022. BidenCash administrators charged a fee for every transaction conducted on the website. The BidenCash marketplace had grown to support over 117,000 customers, facilitated the trafficking of over 15 million payment card numbers and personally identifiable information, and generated over \$17 million in revenue during its operations.

The BidenCash marketplace domains will no longer be operational and will be redirected to a U.S. law enforcement-controlled server, preventing future criminal activity on these sites. The marketplace also sold compromised credentials that could be used to access computers without proper authorization.

Between October 2022 and February 2023, the BidenCash marketplace published 3.3 million individual stolen credit cards for free to promote the use of their services. The stolen data included credit card numbers, expiration dates, Card Verification Value (CVV) numbers, account holder names, addresses, email addresses, and phone numbers.

According to court records, the United States obtained court authorization to seize cryptocurrency funds that BidenCash marketplace used to receive illicit proceeds from its illegal sales.

Erik S. Siebert, U.S. Attorney for the Eastern District of Virginia; John Szydlak, Resident Agent in Charge of the U.S. Secret Service's Frankfurt Resident Office; and Philip Russell, Acting Special Agent in Charge of the FBI Albuquerque Field Office, made the announcement.

This case was investigated by the U.S. Secret Service's Frankfurt Resident Office, the U.S. Secret Service's Cyber Investigative Section, and the FBI Albuquerque Field Office.

The Department of Justice thanks the Dutch National High Tech Crime Unit, The Shadowserver Foundation and Searchlight Cyber for their assistance with the investigation.

The government is represented by Assistant U.S. Attorney Zoe Bedell in these matters.

A copy of this press release is located on the website of the [U.S. Attorney's Office](#) for the Eastern District of Virginia.

Contact

Press Officer
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Updated June 4, 2025



USA's Policy in Action

Building coalitions, strengthening partner capacity, securing global supply chains

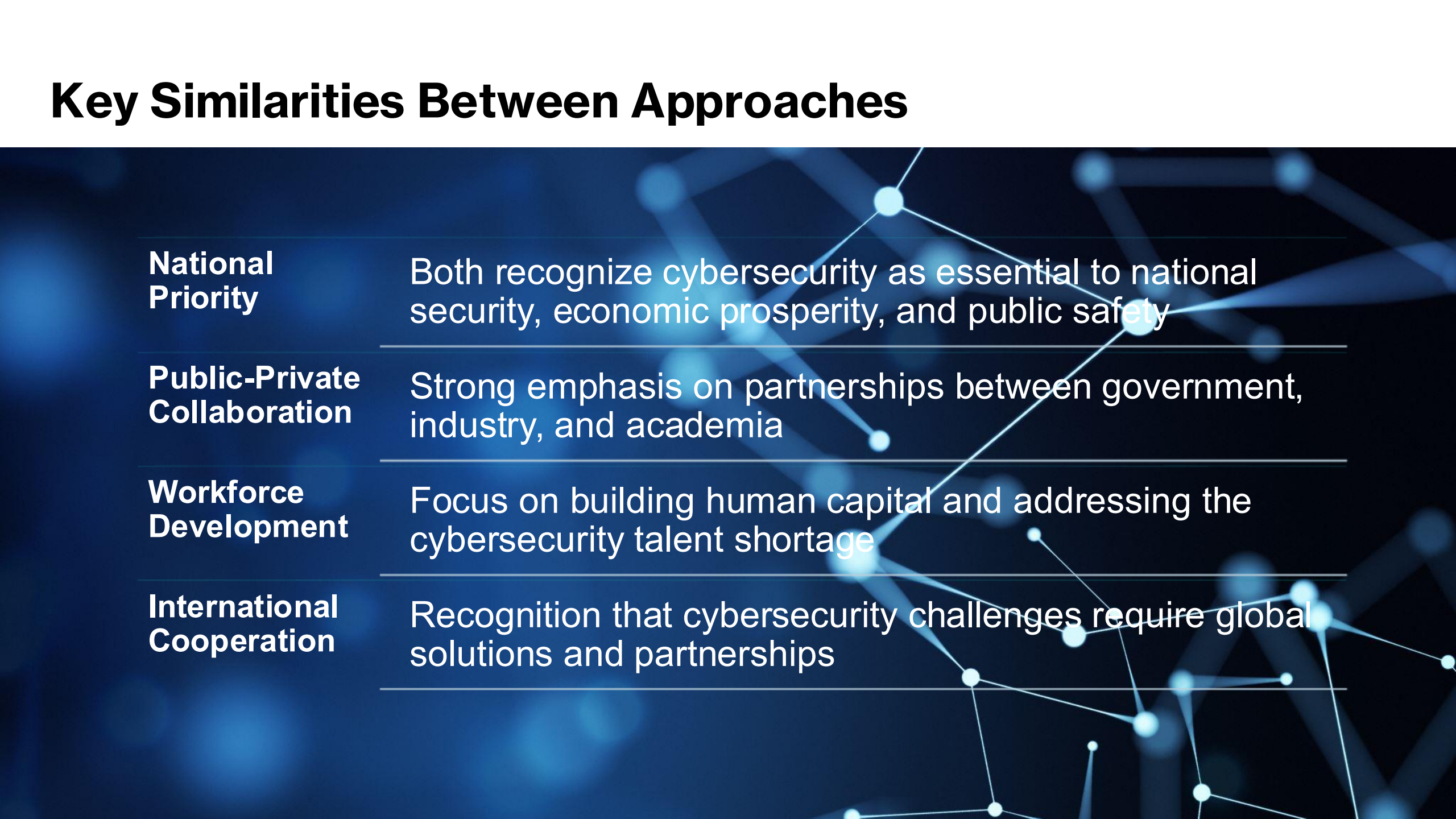


Ukraine

The United States, allies, and partners have invested in Ukrainian cyber capacity building for years, providing a foundation for more immediate assistance in mitigating and recovering from attacks. Before Russia's full-scale invasion of Ukraine, U.S. agencies, including the Federal Bureau of Investigation, U.S. Cyber Command, and the Cybersecurity and Infrastructure Security Agency, shared cyber intelligence with Ukrainian partners. Since the invasion, the United States, United Kingdom, and EU Governments have delivered more than \$100 million in cyber foreign assistance and enabled Ukrainian agencies to access the services of commercial cybersecurity companies. In 2023, the U.S. and nine close partners established the Tallinn Mechanism, a donor coordination group that aims to deliver assistance quickly and efficiently in support of Ukraine's most urgent cybersecurity needs.

<https://2021-2025.state.gov/united-states-international-cyberspace-and-digital-policy-strategy/>

Key Similarities Between Approaches



National Priority	Both recognize cybersecurity as essential to national security, economic prosperity, and public safety
Public-Private Collaboration	Strong emphasis on partnerships between government, industry, and academia
Workforce Development	Focus on building human capital and addressing the cybersecurity talent shortage
International Cooperation	Recognition that cybersecurity challenges require global solutions and partnerships

Key Differences Between Approaches

Europe (ENISA)

Focus on Human Capital

Strong emphasis on competitions, training, and talent development

Standardization

Prioritizes common frameworks across member states

Regulatory Approach

More inclined towards common regulatory frameworks

Workforce Development

Structured competitions (ECSC) and training programs

USA (NCS)

Broader Scope

Covers market forces, liability, R&D, and direct threat disruption

Threat Actor Focus

Explicit strategies for disrupting and dismantling threat actors

Market-Driven

Aims to shape market forces and rebalance incentives

Diverse Approach

R&D, digital identity, and addressing diversity in workforce

Lessons Learned from Both Approaches

Key takeaways that can be applied globally:

Structured
Competitions
Drive Talent

Standardized cybersecurity competitions like ECSC effectively identify and nurture talent while creating valuable networks of specialists

Sustainable
Funding
Models

Government and EU co-funding ensures long-term sustainability of cybersecurity initiatives and consistent talent development

Proactive
Threat
Disruption

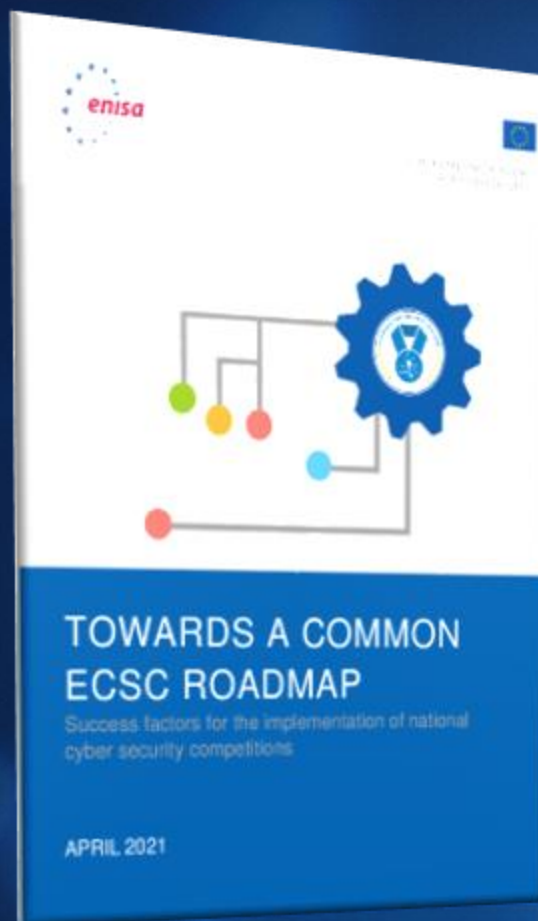
Taking the fight to adversaries through coordinated disruption campaigns can render malicious cyber activity unprofitable

Liability Drives
Security

Shifting liability for insecure software to those best positioned to fix issues can drive systemic improvements in security

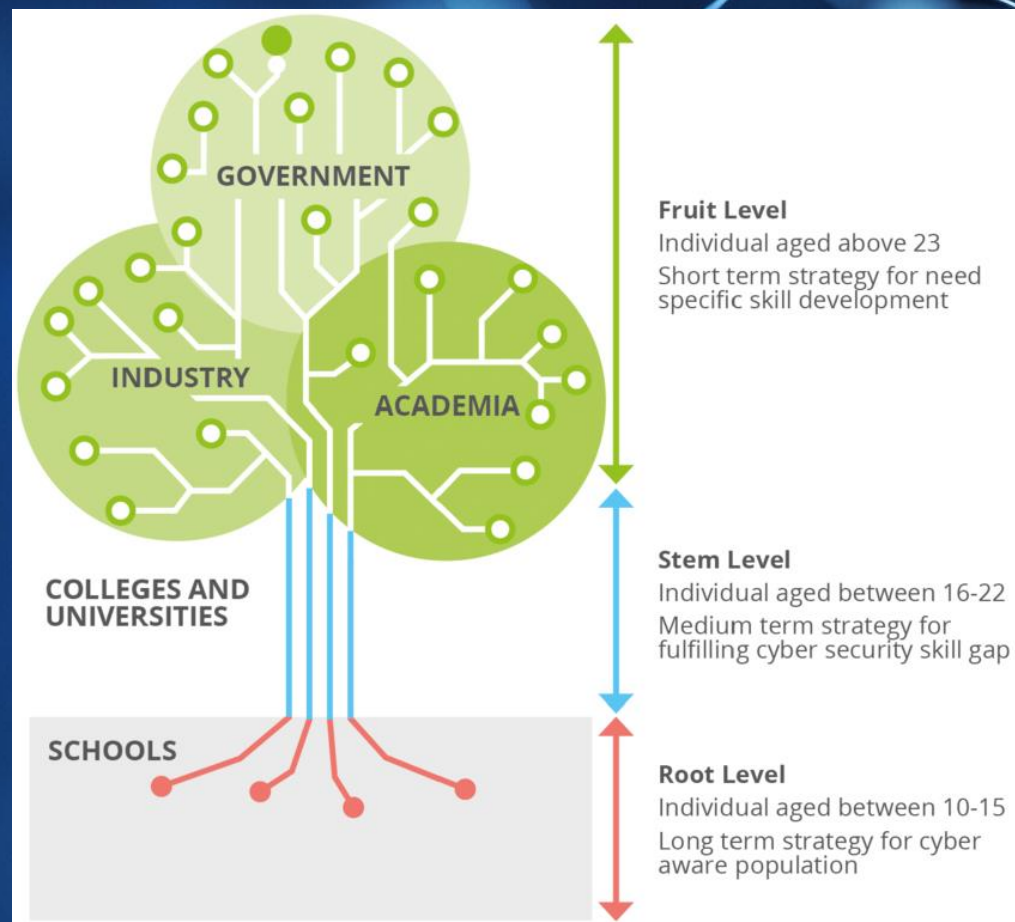
Our Contribution

Analyzed the cyber security ecosystem of 20+ European countries



Tree of Cyber Life

A Tree model to develop a healthy cyber security ecosystem in a country



So, what Europe is Doing right?

In God we trust. All others bring data

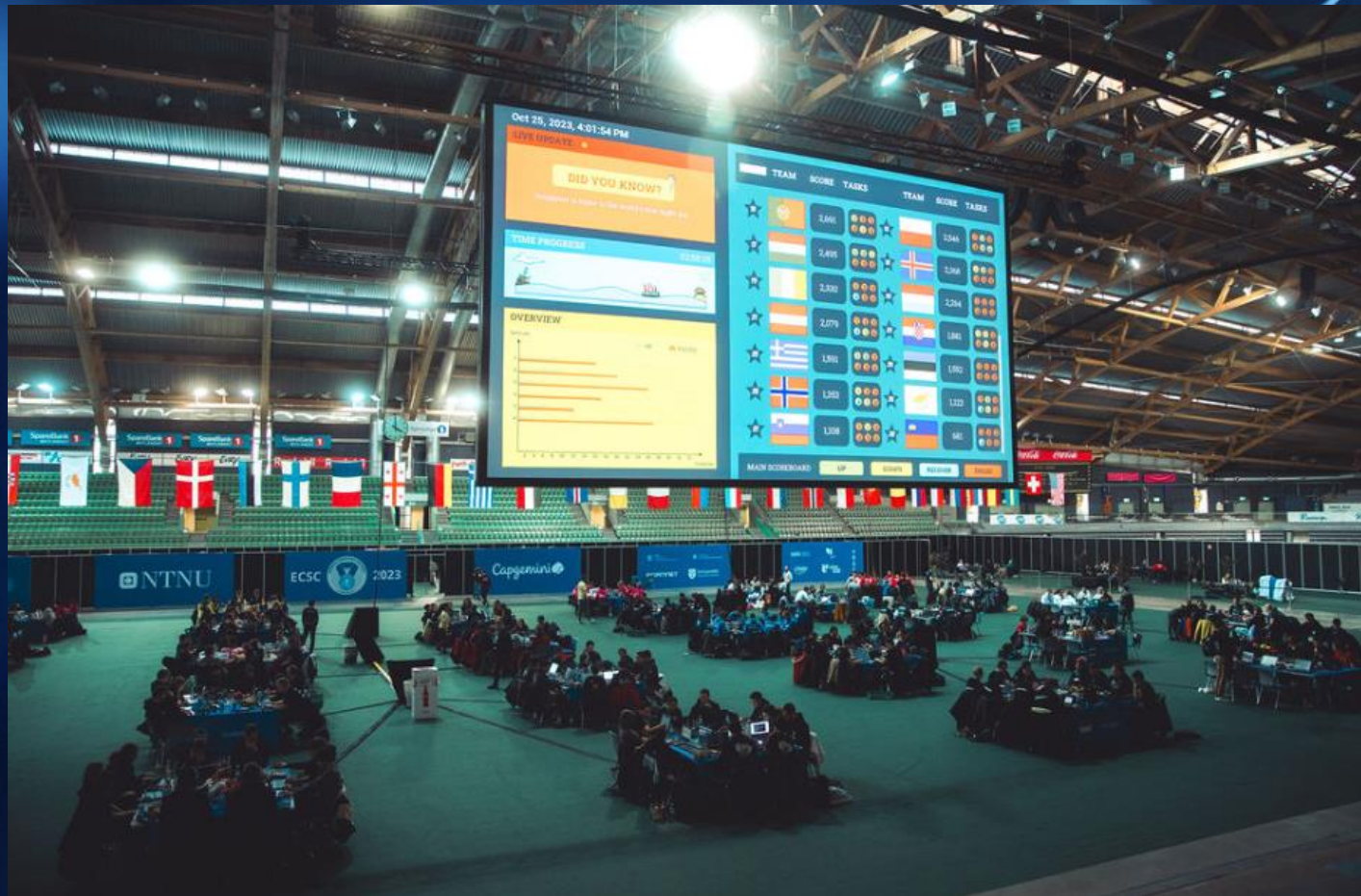
Country/Member State		Participants
Greece		100
Portugal		73
Cyprus		40
Italy		10000
Poland		77
Estonia		80
Hungary		550
Serbia		250
Norway		300
Malta		50
UK		1300
Germany		668
Finland		95
Switzerland		410
Luxembourg		40
ESPAÑA		1000
Albania		60
North Macedonia		
Sweden		10600
Slovenia		21
Kosovo		27
Latvia		400
Croatia		470
Bulgaria		160
Romania		396
Denmark		800
Czechia		11000
Iceland		100
Belgium		850
Netherlands		160
Austria		550
Slovakia		2400
France		3000
Lithuania		450
Ireland		600
Total		47077



Around 50k Europeans Participated in a multinational CTF
Data from ENISA

So, what Europe is Doing right?

People at the center for securing Europe digital future



So, what Europe is Doing right?

In US only 3199 people participated in their national competition in 2021 which is comparable to France

The US has over 334 million people, while France has around 68 million

MEET THE Season IV, US CYBER TEAM®

Over 3,199 athletes from 50 states, Washington, DC, and the Virgin Islands competed in the Season IV, US Cyber Open™ CTF Competition. [View the leaderboard here.](#) After the Open, a select group of 184 athletes was invited to train and compete in the US Cyber Combine™.

The US Cyber Combine is an inclusive, virtual camp focused on athlete training, evaluation, observation, and engagement to provide coaches an opportunity to identify the high-potential athletes to be selected for the Season IV, US Cyber Team or the Pipeline Program.

On Thursday, September 26, 2024, thirty (30) cybersecurity athletes were selected to represent the USA at various global scrimmages and the International Cybersecurity Challenge (ICC) as the Season IV, US Cyber Team. An additional forty-plus (40+) athletes have been invited to join the Pipeline Program.



<https://www.uscybergames.com/season-4-team>

Recommendations for Defcon 33 Audience

Foster Talent Through Structured Programs

- Implement competitions, training, and mentorship programs to identify and develop cybersecurity talent in your organizations

Advocate for Clear Policy and Funding

- Push for consistent policy frameworks and sustainable funding models for cybersecurity initiatives

Promote Public-Private Collaboration

- Build bridges between government, industry, and academia to share intelligence and coordinate responses

Consider Liability for Insecure Products

- Support frameworks that hold software/hardware makers accountable for security while encouraging innovation

Embrace International Partnerships

- Engage in cross-border initiatives and information sharing to strengthen collective defense

As security professionals, you are uniquely positioned to bridge these approaches and drive positive change

Conclusion



Both Europe and the USA have developed distinct but complementary approaches to cybersecurity:

- Europe focuses on human capital development and standardization
- USA emphasizes market forces, threat disruption, and comprehensive resilience



The path forward requires hybrid models and continuous adaptation:

- Combining the strengths of both approaches
- Adapting to evolving threats and technological changes
- Fostering global collaboration while respecting regional differences

Questions & Discussion

Reach out to us if you have any questions and suggestions!



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